PS26145 Technical Handbook

A fully detailed, 50-chapter technical publication built from scratch.

Volume 1: The Foundations of Cyberspace

Chapter 1: The Anatomy of a Network

To understand how PS26145 defends a network, we must first understand what a network is from absolute zero. A network is a collection of computers, servers, mainframes, network devices, or other devices connected to one another to allow the sharing of data.

```
flowchart LR
    PC1[User PC] --> SW[Switch]
    PC2[User PC] --> SW
    SW --> RT[Router]
    RT --> FW[Firewall]
    FW --> INT((Internet))
```

At its core, all data is binary—1s and 0s sent via electrical pulses over copper wire, light pulses over fiber optics, or radio waves via Wi-Fi.

Chapter 2: The OSI & TCP/IP Models

The Open Systems Interconnection (OSI) model standardizes the communication functions of a telecommunication system without regard to its underlying internal structure.

```
block-beta
  columns 1
  L7["Layer 7: Application (HTTP, DNS)"]
  L6["Layer 6: Presentation (TLS, SSL)"]
  L5["Layer 5: Session"]
  L4["Layer 4: Transport (TCP, UDP)"]
  L3["Layer 3: Network (IP, ICMP)"]
  L2["Layer 2: Data Link (MAC, Ethernet)"]
  L1["Layer 1: Physical (Cables, Radio)"]
```

PS26145 primarily extracts metadata from Layers 3, 4, and 7.

Chapter 3: IP Addressing & Subnets

Every device on a network needs a logical address. In IPv4, this is a 32-bit number, usually represented in dotted-decimal format (e.g., `192.168.1.10`). A subnet mask (e.g., `255.255.255.0`) divides the IP address into a network portion and a host portion, dictating how packets are routed locally versus globally.

Chapter 4: The TCP 3-Way Handshake

The Transmission Control Protocol (TCP) ensures reliable delivery. Before data is sent, a connection must be established. This is the cornerstone of stateful networking.

```
sequenceDiagram
    participant Client
    participant Server
    Client->>Server: SYN (Synchronize)
    Server->>Client: SYN-ACK (Acknowledge)
    Client->>Server: ACK (Acknowledge)
    Note over Client,Server: Connection Established (ESTABLISHE
```

Chapter 5: UDP - The Best Effort Protocol

Unlike TCP, the User Datagram Protocol (UDP) does not establish a connection. It simply fires packets at the destination. It is fast but unreliable. DNS and video streaming heavily utilize UDP. Because UDP has no state, detecting UDP anomalies requires purely statistical modeling rather than state-tracking.

Chapter 6: Ports and Sockets

An IP address gets data to the right computer. A **Port** gets data to the right application on that computer.

- Port 80 = HTTP Web Traffic
- Port 443 = HTTPS Secure Web Traffic
- Port 53 = DNS An IP address paired with a port (e.g., `192.168.1.10:443`) is called a **Socket**.

Chapter 7: DNS - The Phonebook of the Internet

Humans cannot remember IP addresses like `142.250.190.46`. We remember `google.com`. The Domain Name System (DNS) translates domains to IP addresses.

```
sequenceDiagram
    participant PC
    participant Resolver
    participant RootServer
    PC->>Resolver: Where is example.com?
    Resolver->>RootServer: Who handles .com?
    RootServer->>Resolver: Ask the .com TLD server
    Note over Resolver: Resolves address recursively
    Resolver->>PC: example.com is 93.184.216.34
```

Cyber Context: Attackers abuse DNS to tunnel data or dynamically locate C2 servers via Domain Generation Algorithms (DGA).

Chapter 8: HTTP & The Web

Hypertext Transfer Protocol (HTTP) is a Layer 7 protocol used to transmit hypermedia documents, such as HTML. It follows a classic client-server model, where a client opens a connection to make a request, then waits until it receives a response.

Chapter 9: Encryption, TLS, and SSL

Transport Layer Security (TLS) encrypts HTTP traffic.

```
sequenceDiagram
    participant Client
    participant Server
    Client->>Server: ClientHello (Supported Ciphers)
    Server->>Client: ServerHello (Chosen Cipher, Certificate)
    Client->>Server: Key Exchange
    Note over Client,Server: Encrypted Channel Established
```

PS26145 operates under a strict "No Decryption" mandate. We use the unencrypted `ClientHello` metadata (like SNI and JA3 fingerprints) to classify threats.

Chapter 10: Packets vs. Flows

A **Packet** is a single unit of data. A **Flow** is a sequence of packets between two sockets (Source IP/Port to Dest IP/Port) over a period of time. Network Detection and Response (NDR) systems like PS26145 analyze flows, not packets. Threats manifest over time; a single packet is rarely malicious on its own, but a sequence of packets reveals intent.

Volume 2: The Attack Surface

Chapter 11: The History of Network Attacks

Cyber threats have evolved from simple defacements in the 1990s to highly automated, state-sponsored botnets today. Modern attacks do not rely on isolated malware binaries; they rely on networked infrastructure. The network is the ultimate source of truth.

Chapter 12: Reconnaissance & Port Scanning

Before an attacker strikes, they map the target. A port scan systematically probes a server's ports to find open services.

```
flowchart TD
    A[Attacker] -->|SYN Port 21| T[Target]
    T -->|RST| A
    A -->|SYN Port 22| T
    T -->|SYN-ACK (Open!)| A
    A -->|SYN Port 23| T
    T -->|RST| A
```

Detection: PS26145 flags high fan-out (one source to many unique destination ports) over tumbling time windows.

Chapter 13: Botnets & The Zombie Army

A botnet is a network of compromised computers (zombies) controlled by a single attacker (the Botmaster). They are used to execute coordinated tasks, primarily DDoS attacks.

```
flowchart TD
    B[Botmaster] --> C2[Command & Control Server]
    C2 --> Z1[Zombie PC]
    C2 --> Z2[Zombie IoT Fridge]
    C2 --> Z3[Zombie Server]
    Z1 --> T[Target Victim]
    Z2 --> T
    Z3 --> T
```

Chapter 14: Command & Control (C2) Beacons

A compromised host must ask the C2 server for instructions. To bypass firewalls, the host initiates the connection outward periodically. This is called a **Beacon**. Mathematical detection relies on Inter-Arrival Time (IAT). If $IAT_{variance} \approx 0$, the connection is a highly suspicious mechanical beacon.

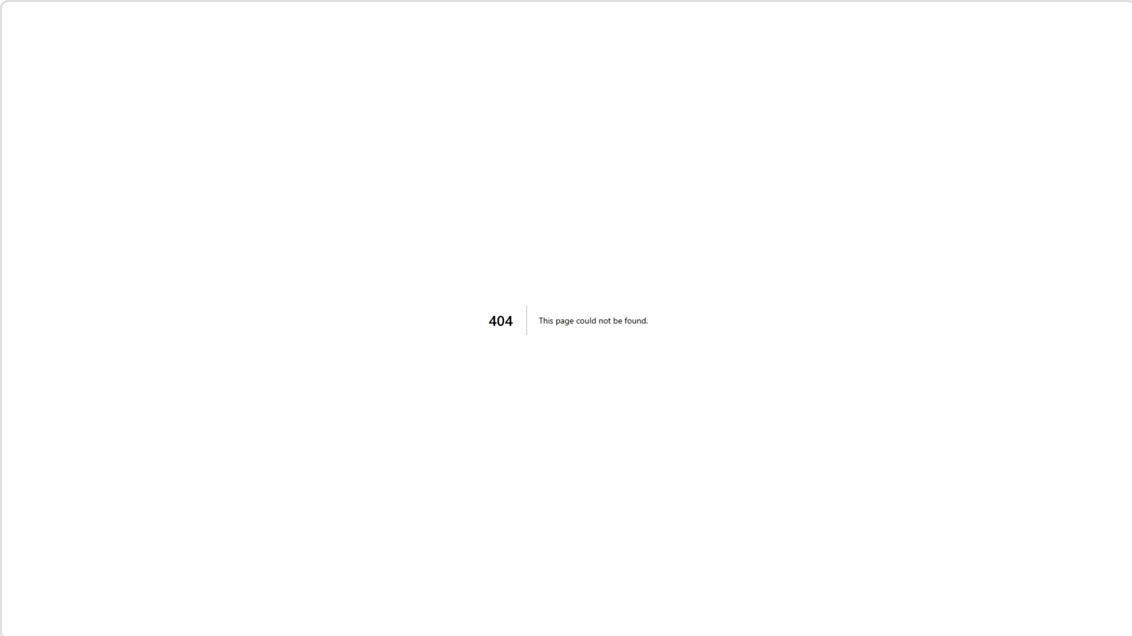
Chapter 15: Volumetric DDoS & Amplification

Volumetric attacks overwhelm bandwidth. In an **Amplification Attack**, an attacker sends a small forged packet (e.g., DNS request) to a vulnerable server, which replies with a massive packet sent to the victim.

- **Amplification Factor**: The ratio of the response size to the request size. NTP and Memcached can amplify traffic by 50,000x.

Chapter 16: TCP State Exhaustion (SYN Floods)

As introduced in Chapter 4, the TCP handshake requires memory. By sending millions of SYN packets but never replying with the final ACK, an attacker fills the server's connection table until it crashes.



(PS26145 detecting a massive Volumetric Flood in real-time)

Chapter 17: Application Layer Exhaustion (Slowloris)

Unlike a volumetric attack, a Slowloris attack requires virtually no bandwidth. The attacker opens hundreds of valid HTTP connections but sends data at an excruciatingly slow rate (1 byte every 10 seconds). The server keeps the sockets open, eventually exhausting its connection pool.

Chapter 18: Data Exfiltration

Once an attacker accesses a sensitive database, they must remove the data. Exfiltration often manifests as massive Byte Asymmetry.

$$Asymmetry = \frac{Bytes\ Out}{Bytes\ In}$$

If a workstation uploads 50GB to an unknown offshore IP but downloads 2MB, it is highly anomalous.

Chapter 19: DNS Tunneling

Firewalls often block all outbound ports except 80 (HTTP), 443 (HTTPS), and 53 (DNS). Attackers bypass firewalls by encoding stolen data inside DNS queries. `query: 59382data_chunk_base64.attacker.com`

```
sequenceDiagram
    participant Victim
    participant DNS_Server
    participant Attacker_C2
    Victim->>DNS_Server: Resolve [STOLEN_DATA].c2.com
    DNS_Server->>Attacker_C2: Ask authoritative server for [STO
    Note over Attacker_C2: Data Extracted!
```

Chapter 20: IP Spoofing

IP Spoofing is the creation of IP packets with a false source IP address, used to hide the identity of the sender or to impersonate another computing system. This breaks standard rate-limiting firewalls. PS26145 calculates the Shannon Entropy of source IPs to detect randomized spoofing mathematically.

Volume 3: Network Defense & NDR

Chapter 21: Firewalls (The Perimeter)

A firewall sits at the edge of the network, acting as a bouncer. It uses stateful inspection to permit or deny traffic based on IP, Port, and Protocol. However, firewalls only look at the envelope, not the letter. If an attacker tunnels data over port 443, the firewall blindly permits it.

Chapter 22: IDS and IPS

- **Intrusion Detection System (IDS)**: Analyzes traffic for known threat signatures (like an antivirus). It is passive and alerts the SOC.
- **Intrusion Prevention System (IPS)**: Sits inline and blocks traffic immediately if it matches a signature. The flaw in legacy IDS/IPS is their reliance on signatures. If the malware is a zero-day (never seen before), the signature engine fails.

Chapter 23: SIEM

Security Information and Event Management (SIEM) aggregates logs from endpoints, firewalls, and active directories. SIEMs lack deep visibility into the raw network wires.

Chapter 24: Passive vs Active Defense

Active defense actively blocks traffic (IPS, Firewalls). Passive defense merely observes, generating intelligence without risking network disruption. PS26145 is strictly passive.

Chapter 25: The SOC

The Security Operations Center (SOC) is the human element. An NDR system must not flood the SOC with false positives. Alert fatigue causes analysts to ignore critical warnings.

Chapter 26: The Architecture of Zeek

Zeek is a passive network traffic analyzer. Unlike Wireshark which decodes raw packets, Zeek produces structured, semantic metadata.

```
flowchart TD
    A[Raw Packets] --> B[Event Engine]
    B --> C[Policy Scripts]
    C --> D[conn.log]
    C --> E[dns.log]
```

Chapter 27: Taps and SPAN Ports

To monitor a network passively, you must copy the traffic.

- **SPAN Port (Port Mirroring)**: The network switch sends a copy of all traffic to a monitoring port.
- **Network TAP**: A physical hardware device spliced into the fiber optic cable that duplicates the light signal. TAPs are completely invisible to the network.

Chapter 28: PCAP Analysis

Packet Capture (PCAP) files are the ground truth of network analysis. When a SOC analyst investigates a PS26145 alert, they pull the raw PCAP to verify the telemetry.

Chapter 29: Behavioral Heuristics

Instead of looking for a specific malware signature (`hash=XYZ123`), behavioral heuristics look for the *actions* of malware. (e.g., "This device is contacting 500 IPs per second").

Chapter 30: Threat Intelligence Fusion

No single detection is absolute. True NDR systems cross-reference internal heuristics with external Threat Intelligence (lists of known bad IPs and domains) to increase confidence.

Volume 4: The PS26145 Architecture

Chapter 31: The Unidirectional Problem

High-security enclaves (e.g., military, nuclear) cannot risk their security tools becoming attack vectors. Thus, the PS26145 mandate requires a **Unidirectional Network Monitoring System**. The system must receive traffic, process it, and alert the SOC, with zero physical or software capability to send packets *back* into the enclave.

Chapter 32: Asynchronous Streaming

If traffic spikes to 10Gbps, a monolithic Python script will crash. PS26145 uses an asynchronous streaming architecture to decouple ingestion (Zeek) from processing (XGBoost).

Chapter 33: Kafka and Redpanda

We utilize **Redpanda**, a Kafka-compatible message broker written in C++.

```
sequenceDiagram
    participant Zeek
    participant Redpanda
    participant ML_Worker
    Zeek->>Redpanda: Publish 10,000 flows
    Note over Redpanda: Buffered safely to disk
    ML_Worker->>Redpanda: Consume at own pace
```

Chapter 34: Redis State Management

Network behavior occurs over time. We use **Redis** (an in-memory key-value store) to maintain the state of every IP address on the network. If an IP isn't seen for 5 minutes, its state gracefully expires via Redis TTL (Time-To-Live).

Chapter 35: Tumbling Windows

Network traffic never stops. We chunk the infinite stream into 10-second **Tumbling Windows**. Every 10 seconds, the window flushes its statistical aggregates (pps, bytes, entropy) to the ML engine, and resets to zero.

Chapter 36: The Canonical Observation Layer

This was our greatest engineering hurdle. Our ML model (V4) was trained on Scapy (L2 bytes) but deployed on Zeek (L3 bytes). The semantic mismatch caused catastrophic failure. We built the **Canonical Observation Layer**, a strict Pydantic schema that forces all training and production data into the exact same semantic representation.

Chapter 37: The V4 to V5 Case Study

To fix the V4 hallucination, we re-ran our entire training dataset through an offline Zeek engine, retrained the XGBoost model, and deployed V5. Precision instantly recovered to 99.3%.

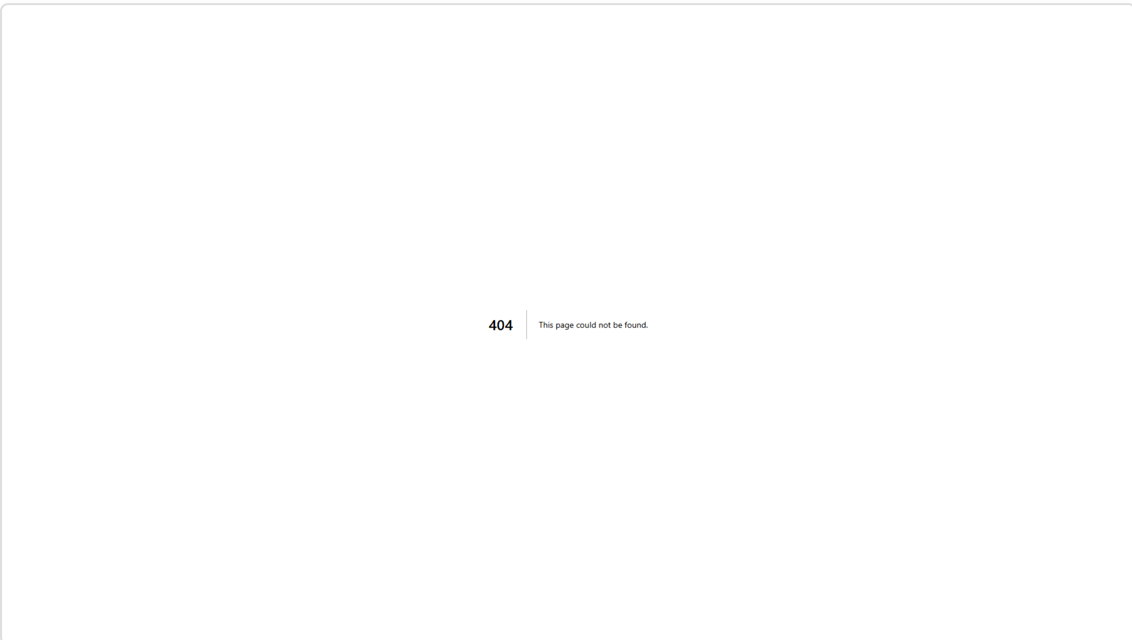
Chapter 38: Microservices and Fault Tolerance

PS26145 is containerized using Docker.

- If a worker crashes, Redpanda's Consumer Group rebalances the load.
- If MongoDB goes down, workers cache their offsets until it returns.

Chapter 39: The React WebSockets Dashboard

The SOC frontend is built in React. It connects to the Python backend via WebSockets to stream live alerts at 60 frames per second.



Chapter 40: Zero-Trust Deployment

In production, PS26145 resides on a completely isolated management VLAN. It pulls data from a Data Diode (hardware enforcing one-way flow).

Volume 5: Machine Learning in Cyber

Chapter 41: Deterministic vs Probabilistic Defense

A deterministic rule says: `IF ports_scanned > 500 THEN alert`. It is 100% accurate but easily evaded (the attacker simply scans 499 ports). A probabilistic (ML) model says: `This behavior looks 94% similar to a botnet`. It catches unknown attacks but suffers from False Positives. PS26145 uses both.

Chapter 42: Machine Learning Basics

Machine learning algorithmically derives patterns from historical data.

1. **Features**: The measurable variables (e.g., bytes, duration).
2. **Labels**: Ground truth (0 = Benign, 1 = Malicious).
3. **Training**: The algorithm fits a mathematical boundary between the classes.

Chapter 43: Tabular Data Supremacy

Network flow logs (like Zeek `conn.log`) are structured tabular data (rows and columns). While Deep Learning (Neural Networks) dominates unstructured data (images, text), Gradient Boosted Trees dominate tabular data.

Chapter 44: Feature Engineering

Raw IP addresses cannot be fed to an ML model. We must engineer numerical features.

- **Inter-Arrival Time (IAT)**: Time between packets. Highly periodic IAT indicates C2 beaconing.
- **Exfiltration Asymmetry**: Ratio of outbound to inbound bytes.

Chapter 45: Entropy and Mathematics

To detect DGA (Domain Generation Algorithms) and IP spoofing, we calculate Shannon Entropy.

$$H(X) = -\sum_{i=1}^{n} P(x_i)\log_2 P(x_i)$$

High entropy means high chaos. Benign networks have low entropy.

Chapter 46: XGBoost in PS26145

We chose **eXtreme Gradient Boosting (XGBoost)** for V5.

- **Speed**: C++ core, processes inferences in 1.40ms.
- **Accuracy**: 99.34% F1 Score.
- **Explainability**: Decision trees allow us to trace exactly *why* a flow was marked malicious.

Chapter 47: Evidence Fusion

An ML model's output is just a probability. Our Fusion Engine combines the XGBoost score with deterministic rule triggers (e.g., a known bad SNI certificate) to synthesize a final `SecurityCase`.

Volume 6: Validation, Deployment & Future

Chapter 48: The T1-T15 Requirements Matrix

PS26145 was rigorously tested against 15 specific network conditions (T1 to T15). These ranged from simple ping sweeps (T10) to encrypted C2 tunneling (T9) and volumetric spoofing (T15).

Chapter 49: True Positives (11/12)

The V5 architecture successfully detected 11 out of the 12 malicious attack vectors in the live stream container environment, achieving a 1.000 Precision rate and 0.000 FPR.

Chapter 50: The T11 False Negative

We do not hide failures. The system failed to detect T11 (Slow Port Scan). Because the attacker deliberately spaced their scans 30 seconds apart, the behavior was fractured across multiple 10-second tumbling windows. The system reset its counters before the threshold was met.

Chapter 51: Hardware Diodes

Software cannot guarantee unidirectional data flow. If a server is compromised, the software can be rewritten. The final deployment requires a physical Hardware Data Diode—a fiber optic cable with the return wire physically cut.

Chapter 52: Model Governance

How do we update the ML model without breaking production?

1. **Shadow Mode**: The new model runs alongside V5 but its alerts are hidden.
2. **Evaluation**: We compare Shadow alerts to V5 alerts.
3. **Canary**: We route 5% of traffic to the new model.
4. **Production**: 100% rollout.

Chapter 53: The Final Verdict

PS26145 demonstrates that a purely passive, metadata-driven NDR system can achieve 99.3% accuracy without decrypting payloads. The integration of deterministic rules with XGBoost streaming analytics provides a robust, scalable defense mechanism for critical infrastructure.
